PROJECT DOCUMENTATION

Team Enrollment Confirmation

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Team Details

Team ID	SWTID-2026-6240
Team Leader Name	SOBIYA S
Team Member 1	SNEHA S
Team Member 2	SATHYAMANI G
Team Member 3	VISHALI S
Department / College	COMPUTER SCIENCE

2. Abstract

This project implements record-level access control in ServiceNow using a Script-Controlled Access Control List (ACL). The application stores team enrollment details, including Team ID, Team Leader, and four Team Members. The ACL evaluates a configured field value and the logged-in user's authorization before allowing an operation such as reading a record. The goal is to ensure that users can access only the records permitted by the defined rule.

3. Introduction

ServiceNow is a cloud-based platform used to build business applications and automate workflows. Access Control Lists help protect application data. An ACL can control access to a table, a record, or a field for operations such as read, create, write, and delete. A scripted ACL adds a condition that is evaluated when a user attempts the protected operation.

4. Problem Statement

When multiple teams use the same enrollment table, every user should not automatically be able to view all team records. Without a suitable access rule, team information may be visible to people who do not need it. This project addresses that problem by checking a field value before granting access.

5. Objectives

- Create a Team Enrollment record structure in ServiceNow.
- Store a unique Team ID, Team Leader, and four member details.
- Configure an ACL for the required table and operation.
- Use a script to evaluate the record's field value and user authorization.
- Test access using permitted and non-permitted users.

- Reduce unintended access to team enrollment information.

6. Scope of the Project

The project focuses on demonstrating read access control for Team Enrollment records. The same approach can be adapted for create, write, or delete operations after defining the required business rules. The project does not automatically create user-to-team relationships; those must be configured using the appropriate group, role, or reference field in the ServiceNow instance.

7. Tools and Technologies

- ServiceNow instance / developer instance
- Access Control List (ACL)
- JavaScript ACL script
- Team Enrollment table
- User and group records for testing

8. Proposed Table Design

Field label	Suggested field type	Purpose
Team ID	String	Identifies the team; for example TEAM-001
Team Leader Name	String or Reference	Stores the team leader
Team Member 1	String or Reference	Stores the first member
Team Member 2	String or Reference	Stores the second member
Team Member 3	String or Reference	Stores the third member
Team Member 4	String or Reference	Stores the fourth member
Enrollment Status	Choice	For example Pending, Confirmed, or Rejected

9. System Workflow

1. An authorized user signs in to ServiceNow. 2.

The user opens a Team Enrollment record.

3. ServiceNow checks the ACL for the requested operation.
4. The ACL script evaluates the record field and the user's authorization. 5. If the

   condition is true, access is granted; otherwise, access is denied.

6. The result is verified using test accounts with different permissions.

10. ACL Configuration Steps

7. Open the ServiceNow application navigator and search for Access Control (ACL).

8. Create an ACL for the Team Enrollment table.
9. Choose the operation to protect, such as Read.
10. Configure any required roles or conditions.
11. Enable the script section and add the approved access rule.
12. Save the ACL and confirm that it applies to the intended table and operation.
13. Test with an authorized account and an account that does not meet the rule.

11. Sample Script Logic

The following is pseudocode to explain the rule. It is not a ready-to-paste production script. Actual field names and the user-to-team mapping must be confirmed in the instance.

```
answer = false;

// Read the team value from the current record.
// Check whether the logged-in user is authorized for that
team. // Grant access only when both checks pass. if
(record_has_valid_team && user_is_authorized_for_that_team) {
answer = true;
}
```

12. Testing and Expected Results

Test case	User condition	Expected result
1	User is authorized for the record's Team ID	Access allowed
2	User is not authorized for that Team ID	Access denied
3	Record has a missing or invalid Team ID	Access denied
4	User tries an operation not covered by permission	Access follows that operation's ACL

13. Security Considerations

- Use the principle of least privilege: grant only the access required.
- Do not rely only on hiding form fields or UI elements; enforce access with ACLs.
- Use the actual user-to-team relationship stored in the instance.
- Test each protected operation separately because read, write, create, and delete are distinct.
- Keep scripts simple, documented, and reviewed before deployment.

14. Advantages

- Restricts access based on a defined record value.

- Helps protect team enrollment information.
- Supports consistent access decisions through the platform's ACL mechanism.
- Can be extended to additional teams and operations.
- Provides a clear method for testing authorized and unauthorized access.

15. Limitations

- The ACL works only when configured for the correct table and operation.
- Incorrect field names or user-to-team mapping can cause unexpected access results.
- Additional ACLs may also apply, so the complete access configuration must be tested.
- Sample scripts require adaptation to the actual ServiceNow data model.

16. Future Enhancement

Future versions may use reference fields for users and teams, add an approval workflow for enrollment, record access attempts for auditing, and provide separate permissions for team leaders and members.

17. Conclusion

This project demonstrates how a Script-Controlled ACL can restrict access to Team Enrollment records based on a field value and the logged-in user's authorization. It provides a practical example of record-level security in ServiceNow. Proper configuration and testing help ensure that only permitted users can access the relevant team information.

— End of Project Documentation —